Assignment_2

Task: 1

From your Assignment 1 brief, list every distinct job the system does — at least six Capabilities (the verbs). Submit — the capability list in design.pdf.

Title: Capabilities

Here are six distinct jobs the CampusEats system does:

1. **Browse Menu:** Users can view available food items from Vendors.
2. **Place Orders:** Users submit an order for selected items.
3. **Deliver Food:** Delivery staff fulfil and transport orders.
4. **Make Payments:** Users Pay for their order Securely.
5. **Track Order:** User monitor the status of their orders in real time.
6. **Manage Vendors:** Vendors update Menus, Prices, and availability.

Task: 3

For each service, write a contract: a table of the operations other services may call — operation name, input, output, errors. Do not mention tables, id formats, or SQL; expose the operation, hide the inside. Submit — the contract tables in design.pdf (≥3 operations for your two main services, ≥1 for the rest).

Service Contracts

1. **Menu Service**

Operation	Input	Output	Errors
getMenu	vendorID	menuList	vendorNotFound

2. **Order Service**

Operation	Input	Output	Errors
placeOrder	Order Details	confirmation ID	itemNotFound, paymentFailed
getOrder	orderID	order details	orderNotFound

3. **Delivery Service**

Operation	Input	Output	Errors
assignDelivery	orderID	deliveryID	noDriverAvailable
trackDelivery	orderID	deliveryStatus	orderNotFound

4. **Payment Service**

Operation	Input	Output	Errors
processPayment	Payment Details	Payment Status	insufficientFunds, invalidCard
refundPayment	orderID	refund status	refundNotAllowed, orderNotFound

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5. Vendor Management Service

Operation	Input	Output	Errors
updateMenu	vendorID, items	confirmation	vendorNotFound, invalidItem
manageVendor	Vendor Details	Vendor Status	duplicateVendor, invalidData

6. Tracking Service

Operation	Input	Output	Errors
tackOrder	orderID	orderStatus	orderNotFound

Task: 4

Specify your central operation — placeOrder — in full: its inputs, outputs, error cases, and the internal details it must hide from callers.

Submit — the full specification in design.pdf.

Central Operation- PlaceOrder

Category	Details
Inputs	• userID- Identifies the Users placing the order • vendorID- Identifies the food Vendor. • items[]- List of menu items with quantity. • paymentDetails- card/UPI/Wallet info. • deliveryAddress- Location for delivery.
Outputs	• orderID- Unique identifier for the order. • confirmationStatus- success/failure. • estimatedDeliveryTime- Calculated time window.
Error Cases	• itemNotFound- Requested item not on Menu. • vendorUnavailable- Vendor closed or Offline. • paymentFailed- Insufficient funds or invalid payment method. • deliveryUnavailable- No delivery staff available.
Internal Details (Hidden)	• Validate Menu items against vendor's catalogue. • Check vendor availability and stock. • Reserve items in **inventory**. • Process payment securely. • Create order record in **Orders** database. • Notify Delivery Service to assign Driver. • Return confirmation to User.

Task: 6

Check each service against the five properties of a Service (reachable, self-contained, has a contract, independent, loosely coupled). Where one fails, say so and note how you would fix it.

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Submit — the validation table in design.pdf.

Service	Reachable	Self-Contained	Has a contact	Independent	Loosely Coupled	Notes/Fixes
Menu Service	Yes	Yes	Yes	Yes	Yes	Fully Valid
Order Service	Yes	Yes	Yes	Yes	Yes	Central Service, Contract defined
Delivery Service	Yes	Yes	Yes	Yes	Yes	Valid, Ensure Driver Assignment doesn't directly access Order DB
Payments Service	Yes	Yes	Yes	Yes	Yes	Valid, Must only expose processPayment and refundPayment, Payment must not expose card details
Vendor Management	Yes	Yes	Yes	Yes	Yes	Valid, Ensure Vendors do not share table with Menu Service
Tacking Service	Yes	Yes	Yes	Yes	Yes	Valid, relies only on OrderID reference